

CELL COMMUNICATION

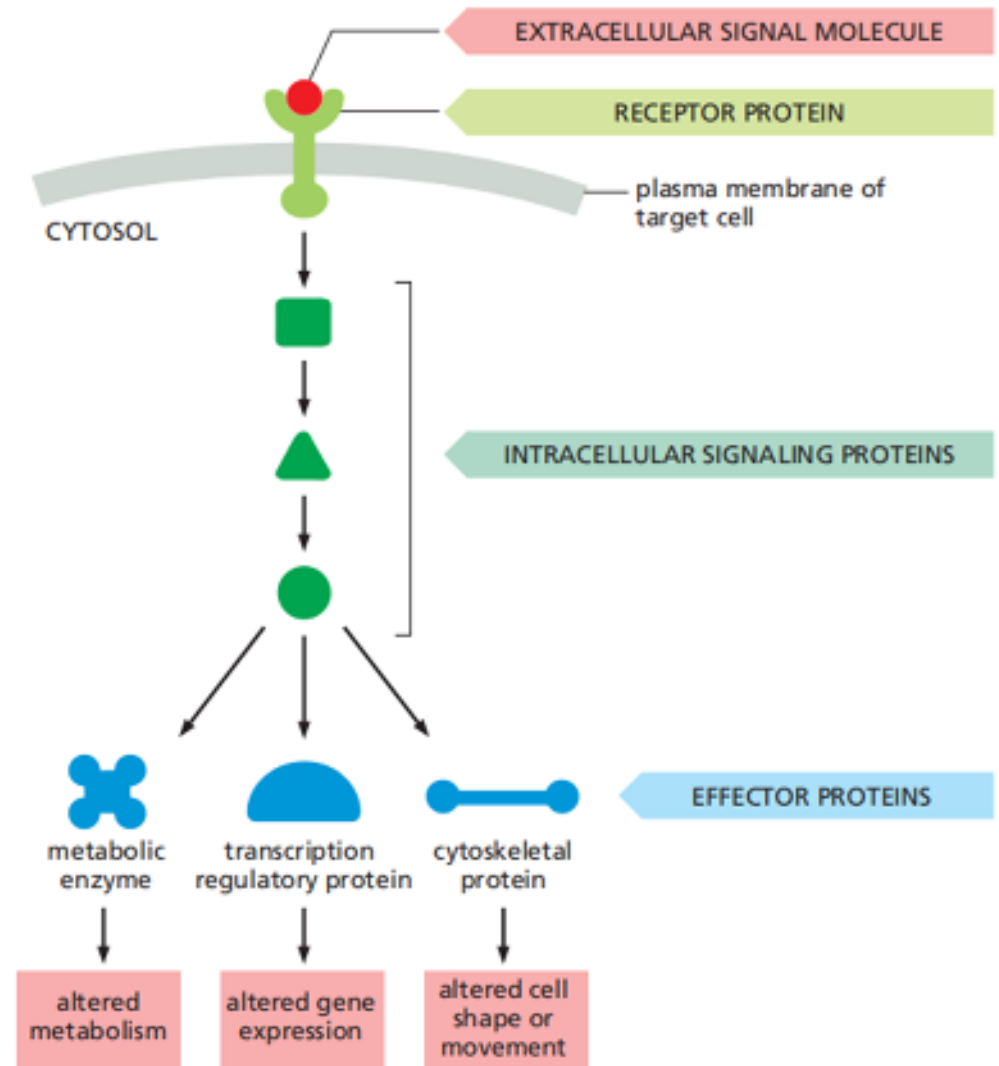
Sri Wulandari

Introduction

- Cells of multicellular organisms detect and respond to internal and extracellular signals that control their growth, division, and differentiation during development, as well as their behavior in adult tissues → cell communication
- **Extracellular signal molecules** bind to receptors to mediate cell communication
- Extracellular molecules trigger **intracellular signaling pathways**
- Intracellular signaling pathways induce **signal transduction pathway**

Signal Transduction Pathways

- A **signal transduction pathway** is a series of steps by which a signal on a cell's surface is converted into a specific cellular response
- The molecules that relay a signal from receptor to response are mostly **proteins**
- Like falling dominoes, the receptor activates another protein, which activates another, and so on, until the protein producing the response is activated
- At each step, the signal is **transduced** into a different form, usually a conformational change



Types of Chemical Signaling

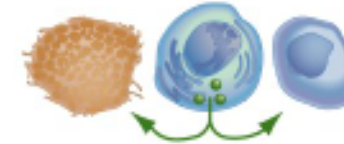
Chemical signaling between cells is one of the most important ways that activities of tissues and organs are coordinated.

The nervous system is the other major coordinating system in animals, but even here chemical signaling is used between adjacent neurons.

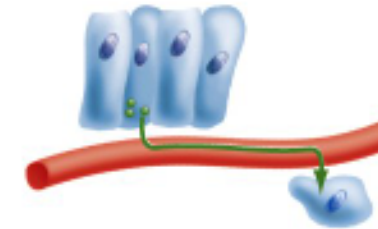
Autocrine signals act on the same cell that secretes them.



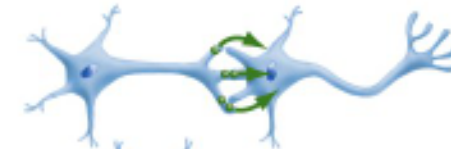
Paracrine signals diffuse locally and act on neighboring cells.



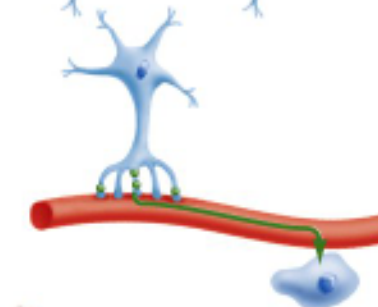
Endocrine signals are hormones carried between cells by blood or other body fluids.



Neural signals diffuse a short distance between neurons.



Neuroendocrine signals are released from neurons but act on distant cells.



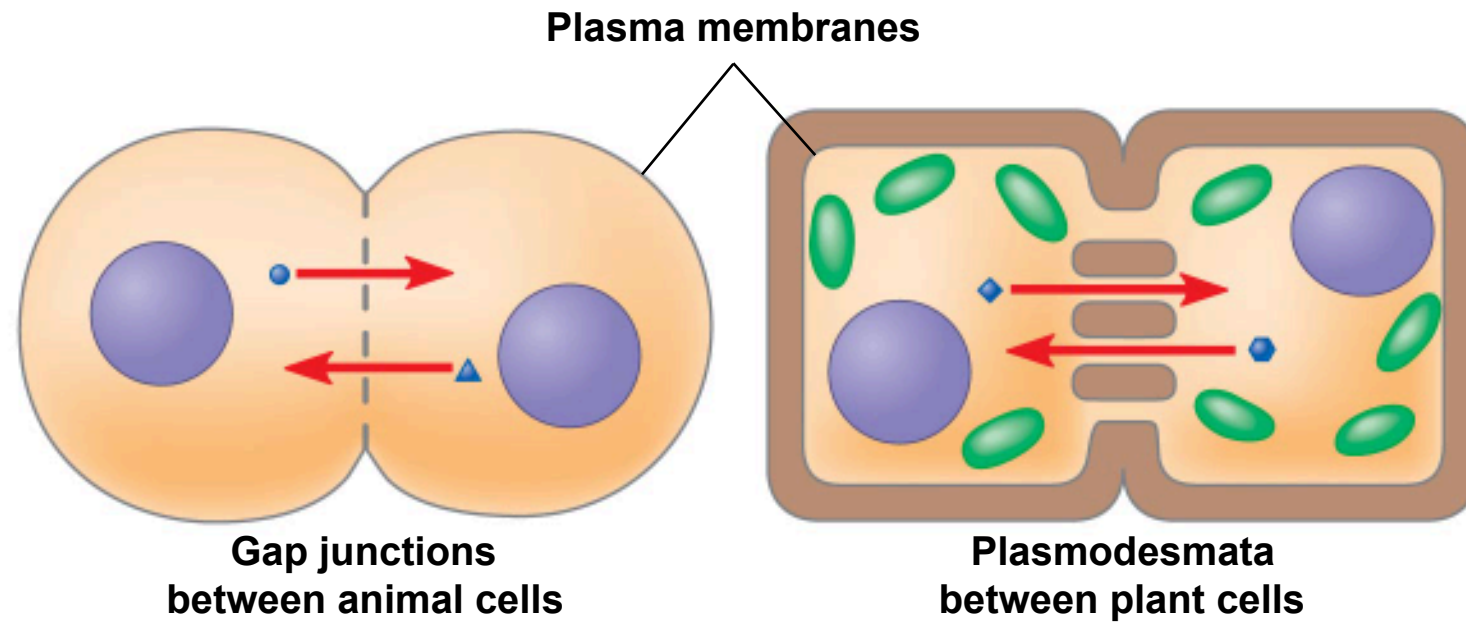
Pheromones are released into the environment and act on a different individual.



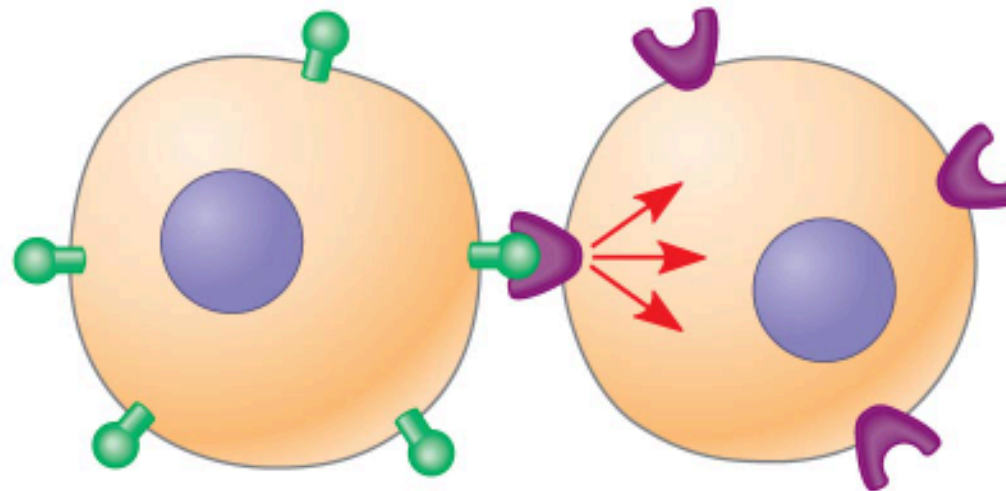
Figure 47-1 Biological Science, 2/e
© 2005 Pearson Prentice Hall, Inc.

Modes of cell signaling

1. **Direct** cell-cell or cell-matrix
2. **Indirect: Secreted molecules.**
 - A. Endocrine signaling. The signaling molecules are hormones secreted by endocrine cells and carried through the circulation system to act on target cells at distant body sites.
 - B. Paracrine signaling. The signaling molecules released by one cell act on neighboring target cells (neurotransmitters).
 - C. Autocrine signaling. Cells respond to signaling molecules that they themselves produce (response of the immune system to foreign antigens, and cancer cells).



(a) Cell junctions

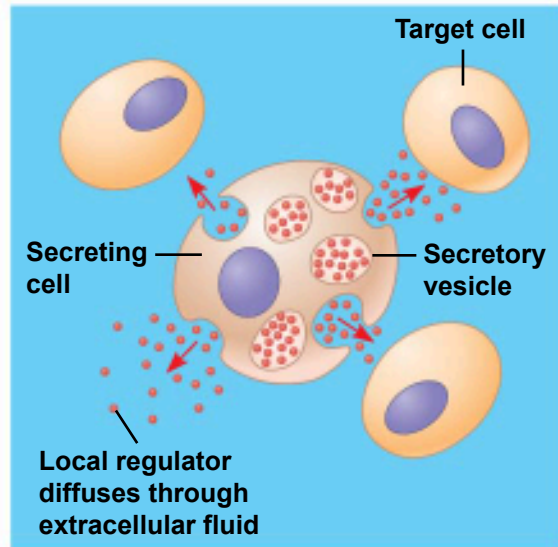


(b) Cell-cell recognition

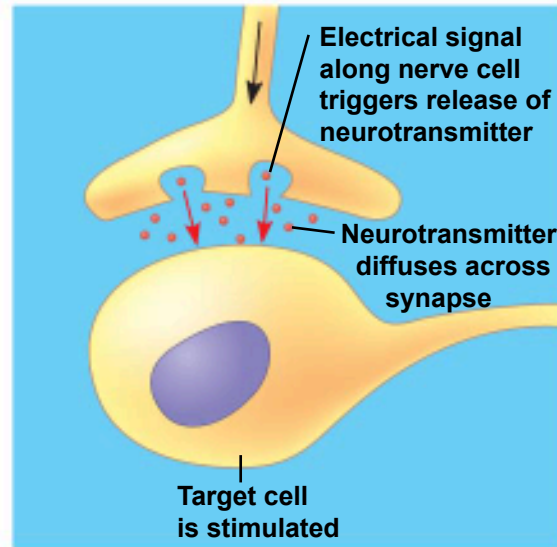
Cell signaling: Local and Long-distance signaling

- Cells in a multicellular organisms communicate by chemical messengers
- Animal and plant cells have cell junctions that directly connect the cytoplasm of adjacent cells
- In local signaling, animal cells may communicate by direct contact
- In many other cases, animal cells communicate using local regulators, messenger molecules that travel only short distances
- In long-distance signaling, plants and animals use chemicals called hormones

Local signaling

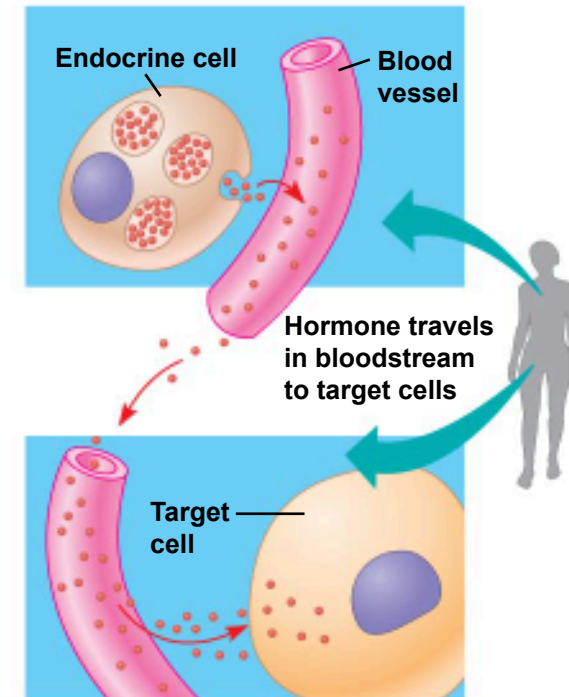


(a) Paracrine signaling



(b) Synaptic signaling

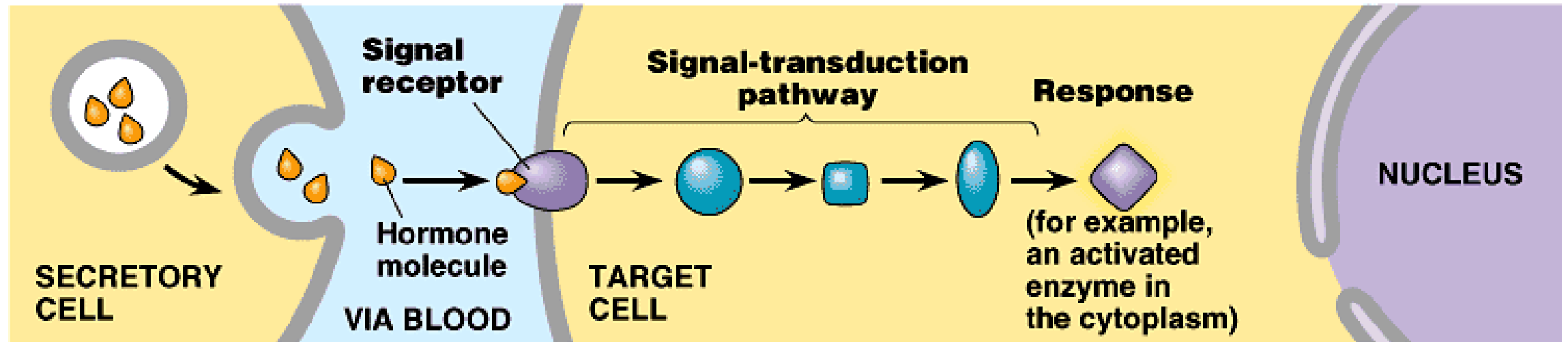
Long-distance signaling



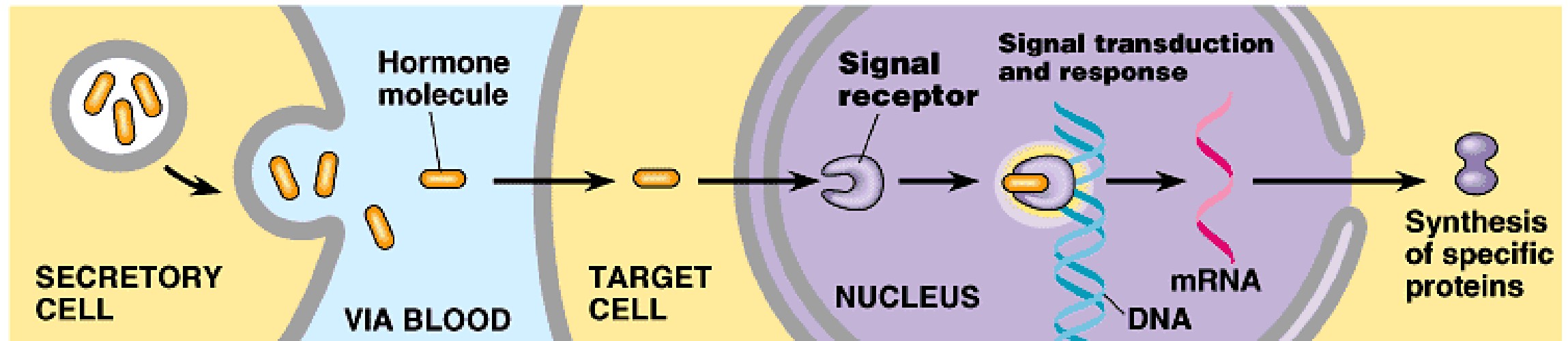
(c) Hormonal signaling

Stages of Cell Signaling

- Cells receiving signals went through three processes:
 - **Reception**
 - **Transduction**
 - **Response**
- Target organ specificity is the result of specific receptor molecules for the hormone, either on the plasma membrane surface, or in some cases in the cytoplasm, of cells in the target organ.



(a) Receptor in plasma membrane



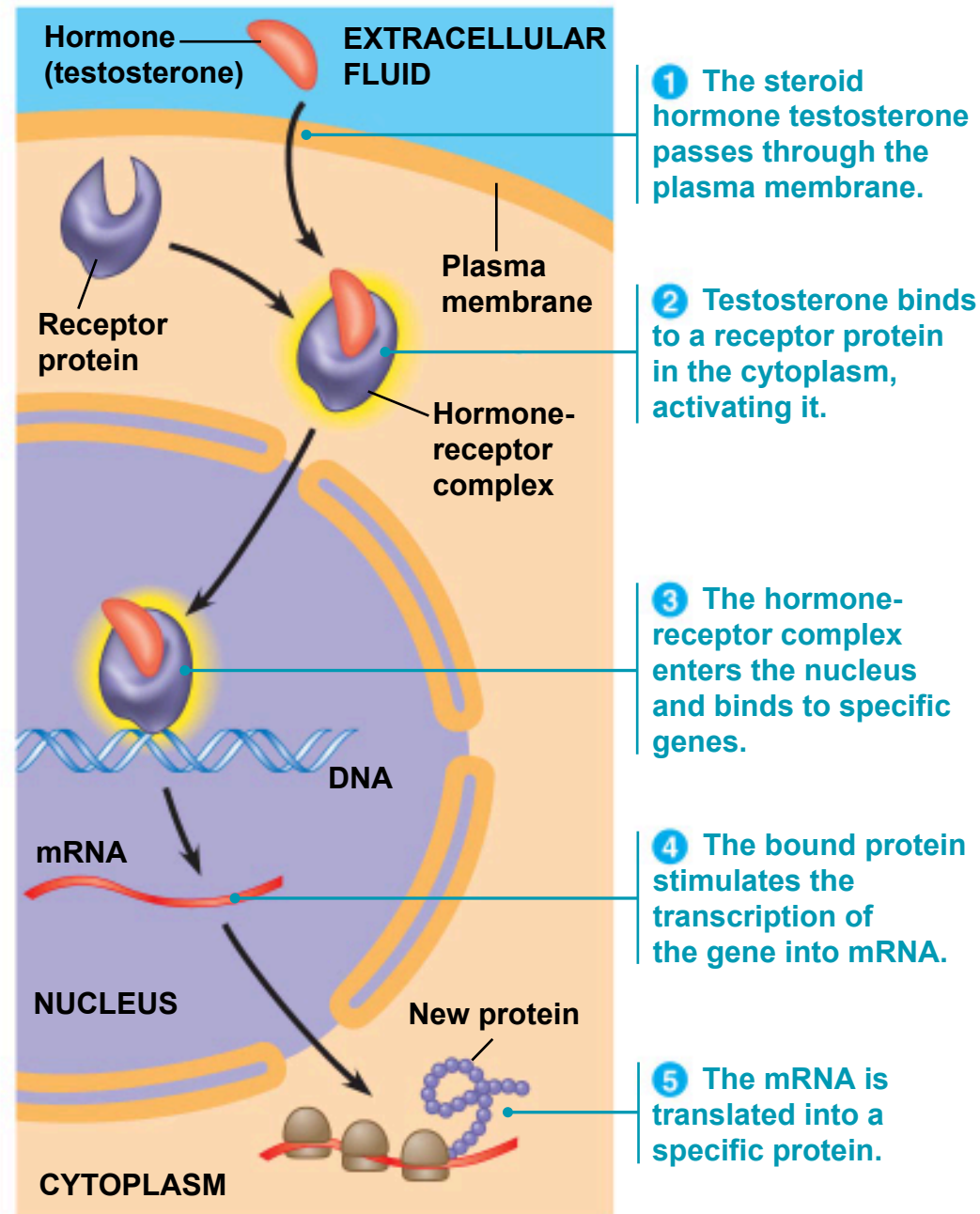
(b) Receptor in cell nucleus

RECEPTION

- A signal molecule binds to a receptor protein, causing it to change shape
- The binding between a signal molecule (ligand) and receptor is highly specific
- A conformational change in a receptor is often the initial transduction of the signal
- Most signal receptors are plasma membrane proteins

Intracellular Receptors

- Some receptor proteins are intracellular, found in the cytosol or nucleus of target cells
- Small or hydrophobic chemical messengers can readily cross the membrane and activate receptors
- Examples of hydrophobic messengers are the steroid and thyroid hormones of animals
- An activated hormone-receptor complex can act as a transcription factor, turning on specific genes



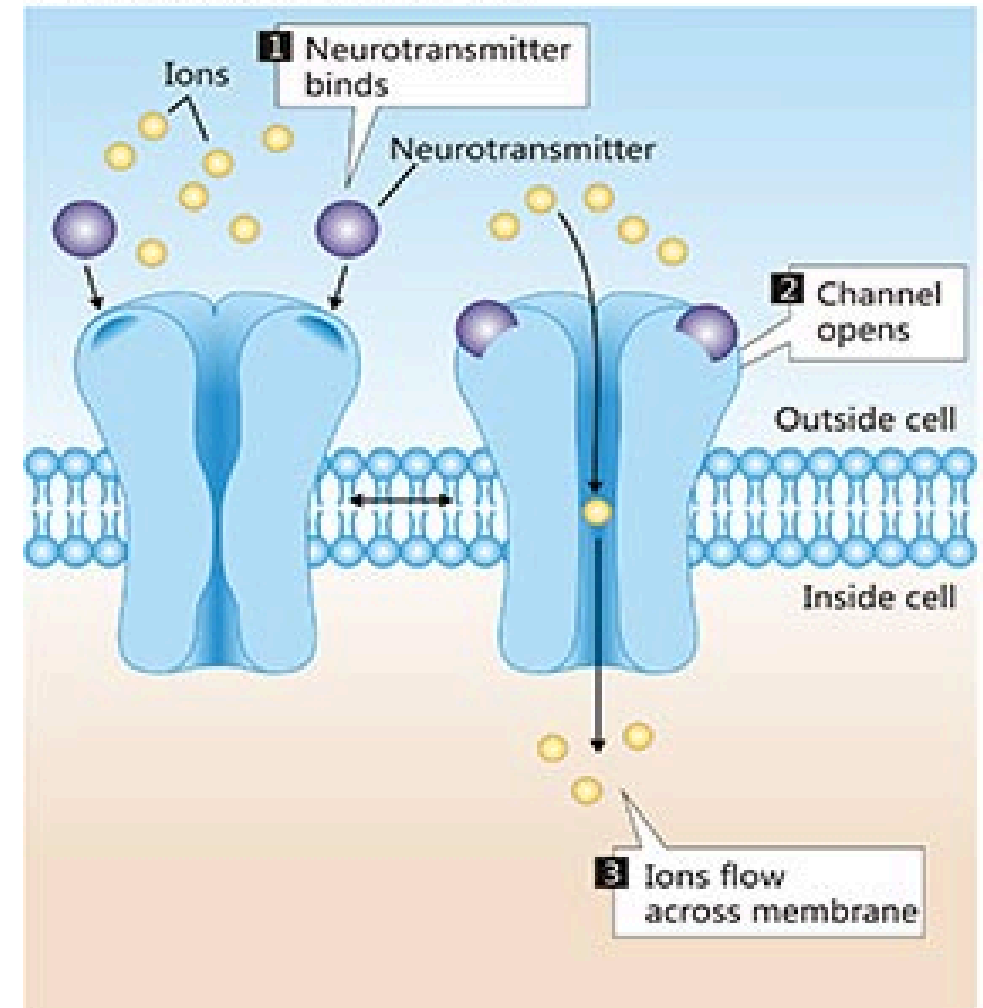
Receptors in the Plasma Membrane

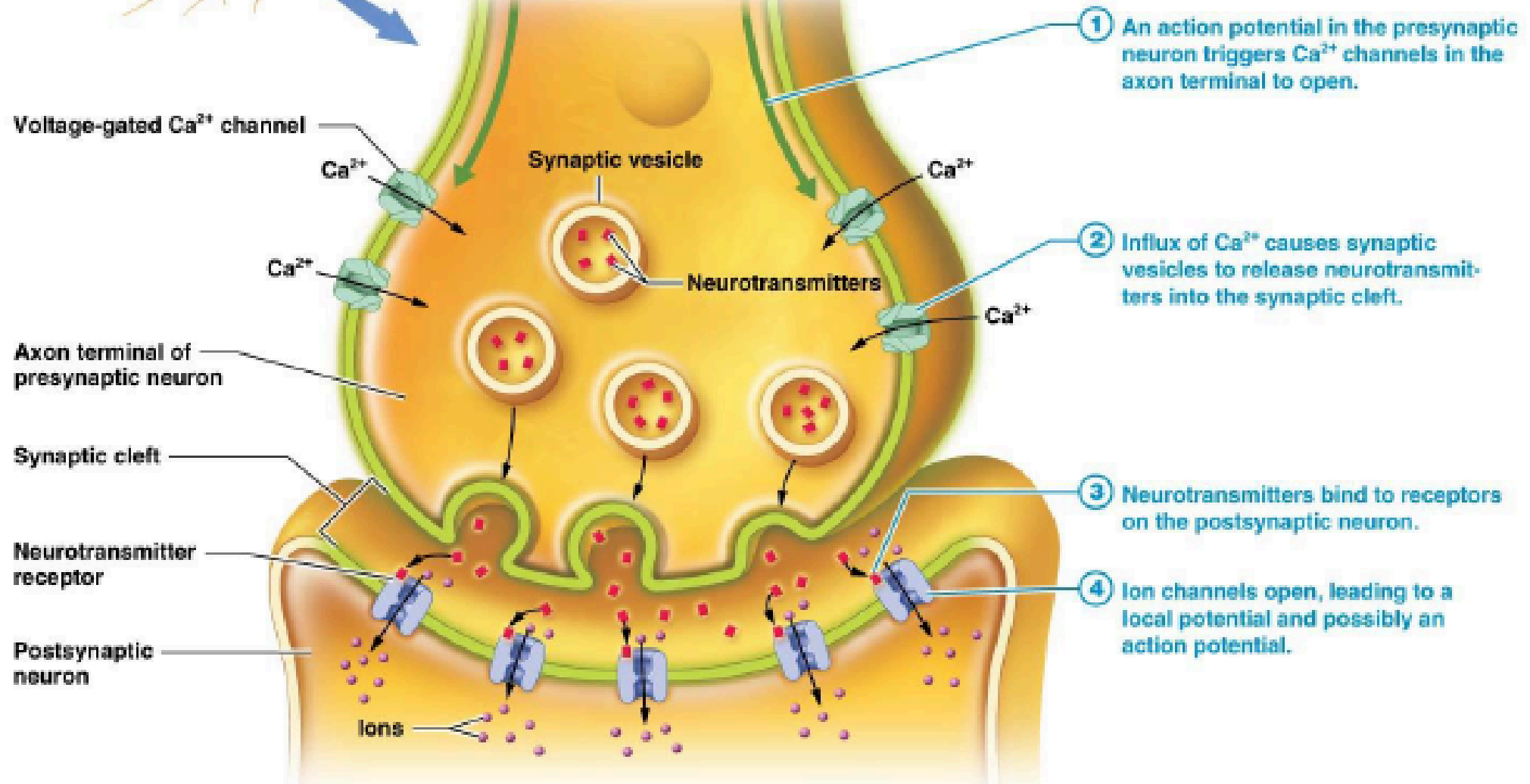
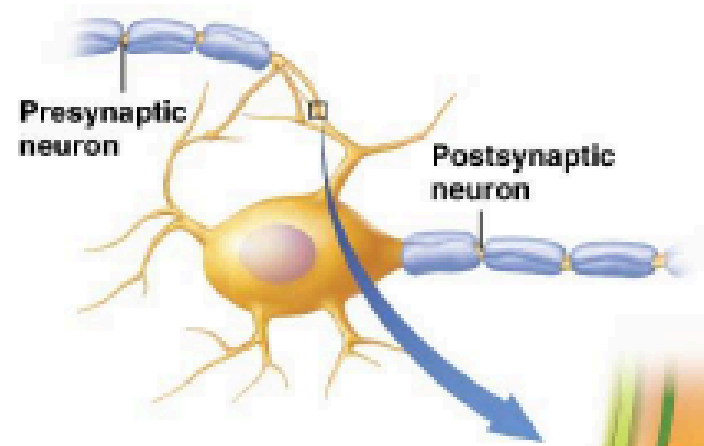
- Most water-soluble signal molecules bind to specific sites on receptor proteins in the plasma membrane
- There are three main types of membrane receptors:
 - Ion channel receptors
 - G-protein-linked receptors
 - Enzyme-coupled receptor

Ion channel receptors

- An ion channel receptor act as a gate when receptor change shape
- When a signal molecule binds as a ligand to the receptor, the gate allows specific ions, such as Na^+ , K^+ or Ca^{2+} through the channel
- Example: neuromuscular junction ion channel

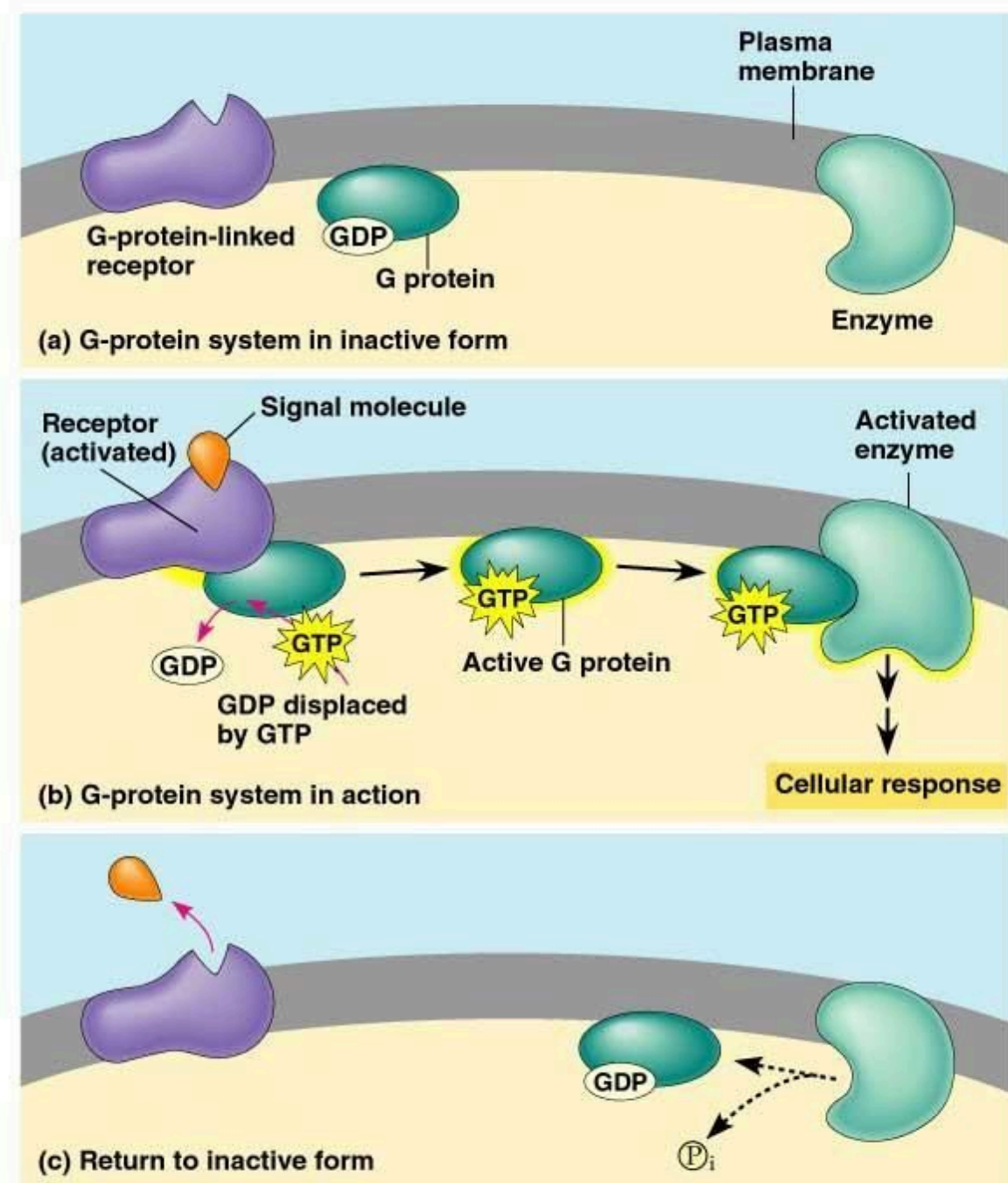
(A) Ligand-gated ion channels

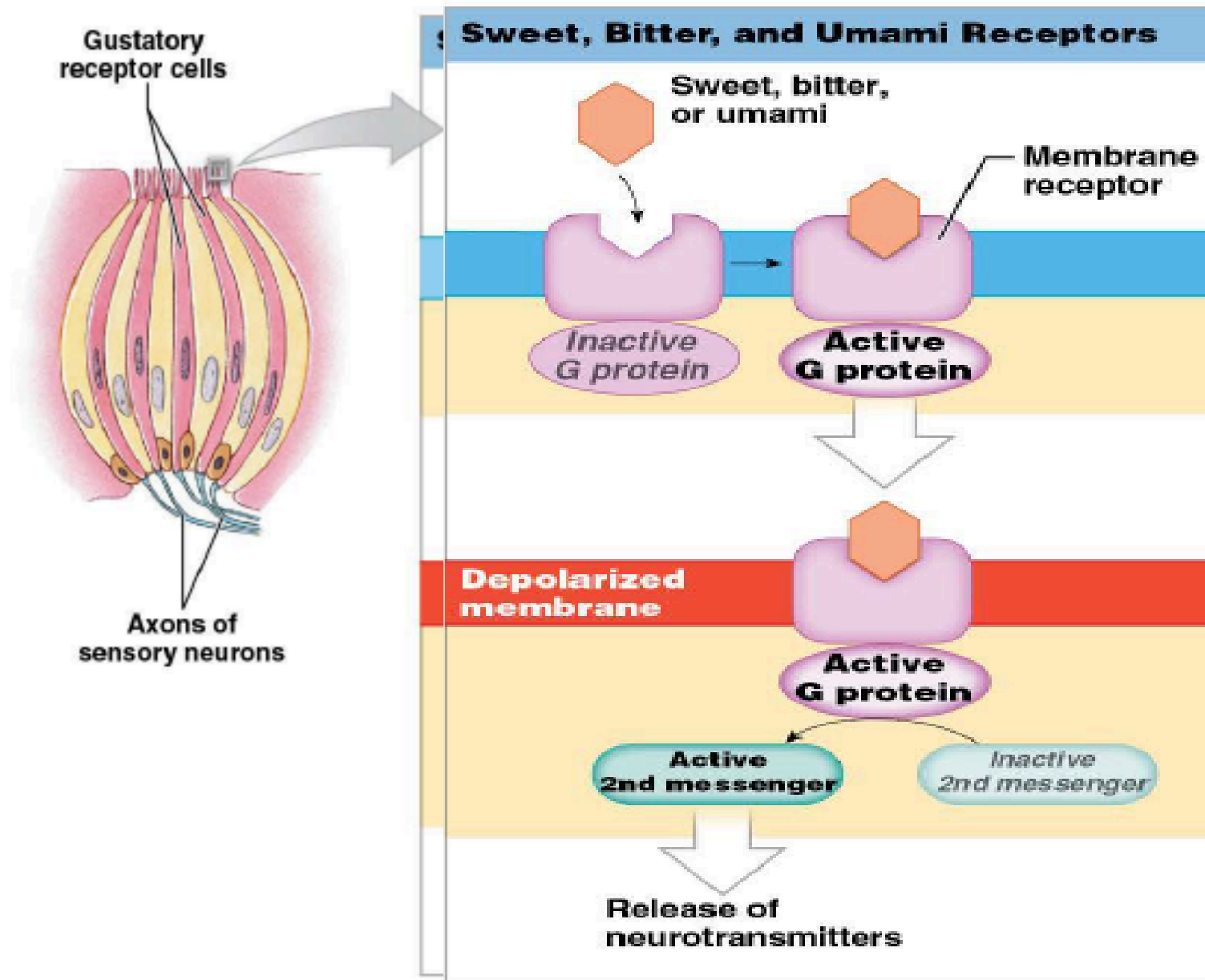




G-protein-linked receptors

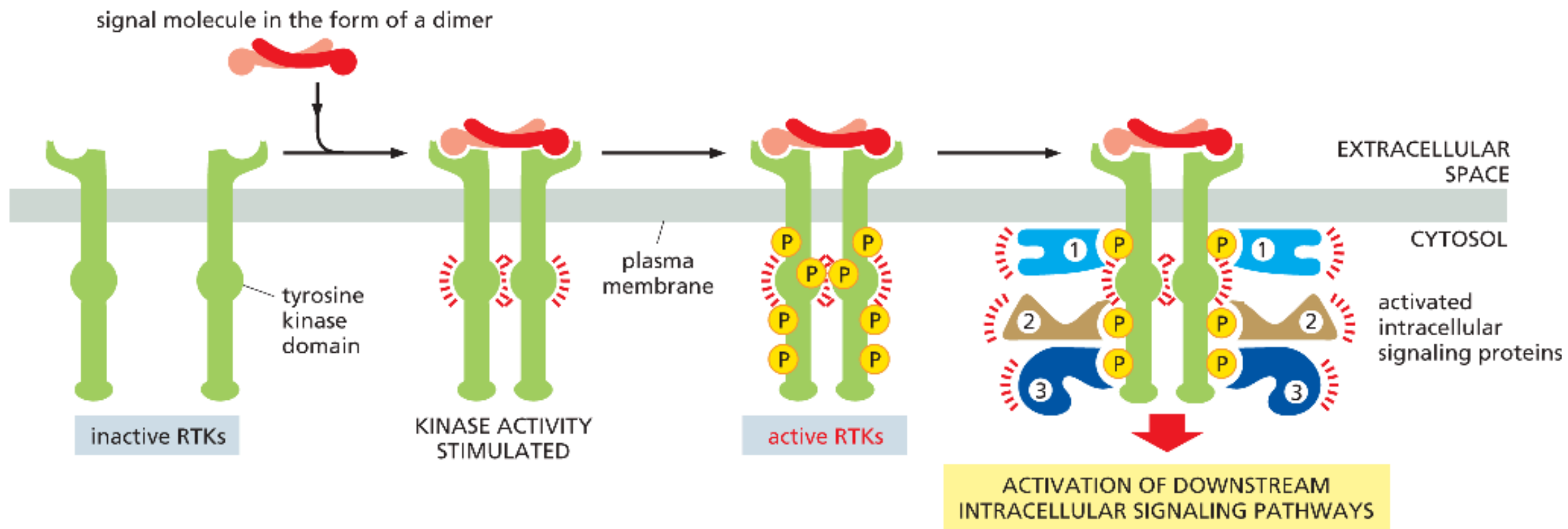
- A G-protein-linked receptor is a plasma receptor that work with the help of a G protein
- The G –protein acts as an on/off switch: if GDP bound to the G protein, the G protein is inactive





Enzyme-coupled receptor

- the cytoplasmic domain of the receptor either acts as an enzyme itself or forms a complex with another protein that acts as an enzyme
- Example : receptor tyrosine kinase (RTK)
- RTK are membrane protein that attach phosphate to tyrosine
- RTKs bind growth factors that regulate cell differentiation, proliferation, survival, metabolism, and migration

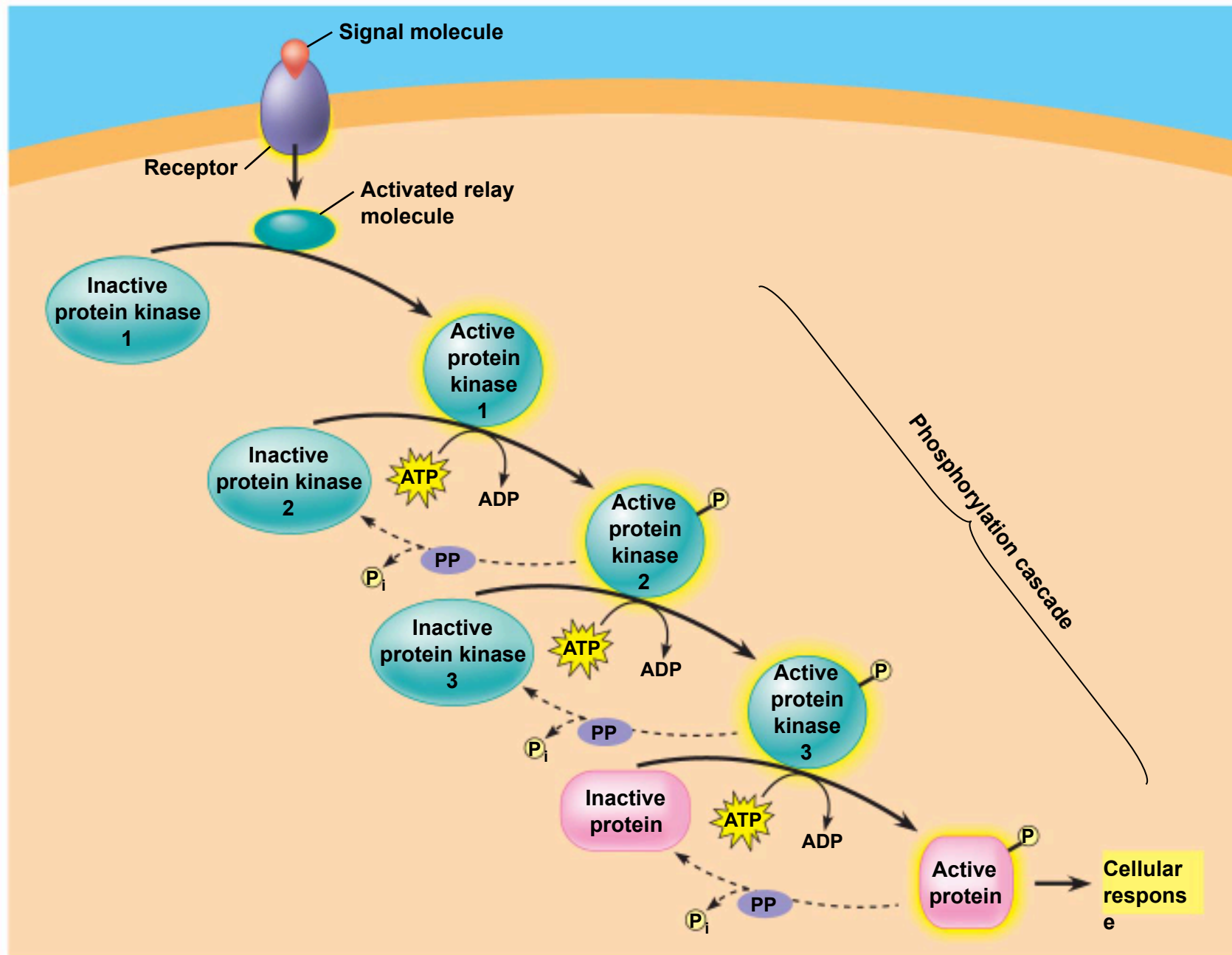


TRANSDUCTION

- Cascades of molecular interactions relay signals from receptors to target molecules in the cell
- Transduction usually involves multiple steps
- Multistep pathways can amplify a signal: A few molecules can produce a large cellular response
- Multistep pathways provide more opportunities for coordination and regulation

Protein Phosphorylation and Dephosphorylation

- Protein phosphorylation is the addition of a phosphate group to a protein, usually an amino acid, by a protein kinase, which alters the protein's 3D structure and function.
 - In many pathways, the signal is transmitted by a cascade of protein phosphorylations
- Protein dephosphorylation is the reverse process, where the phosphate group was removed, returning the protein to its original state.
 - Phosphatase enzymes remove the phosphates
- This phosphorylation (kinases) and dephosphorylation (phosphatases) system acts as a molecular switch, turning activities on and off

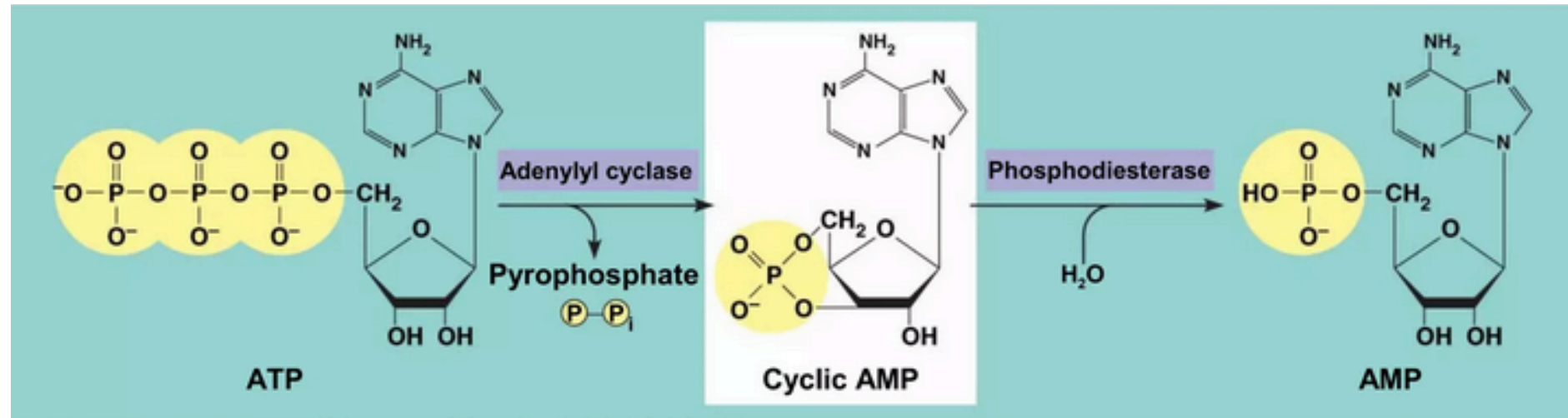


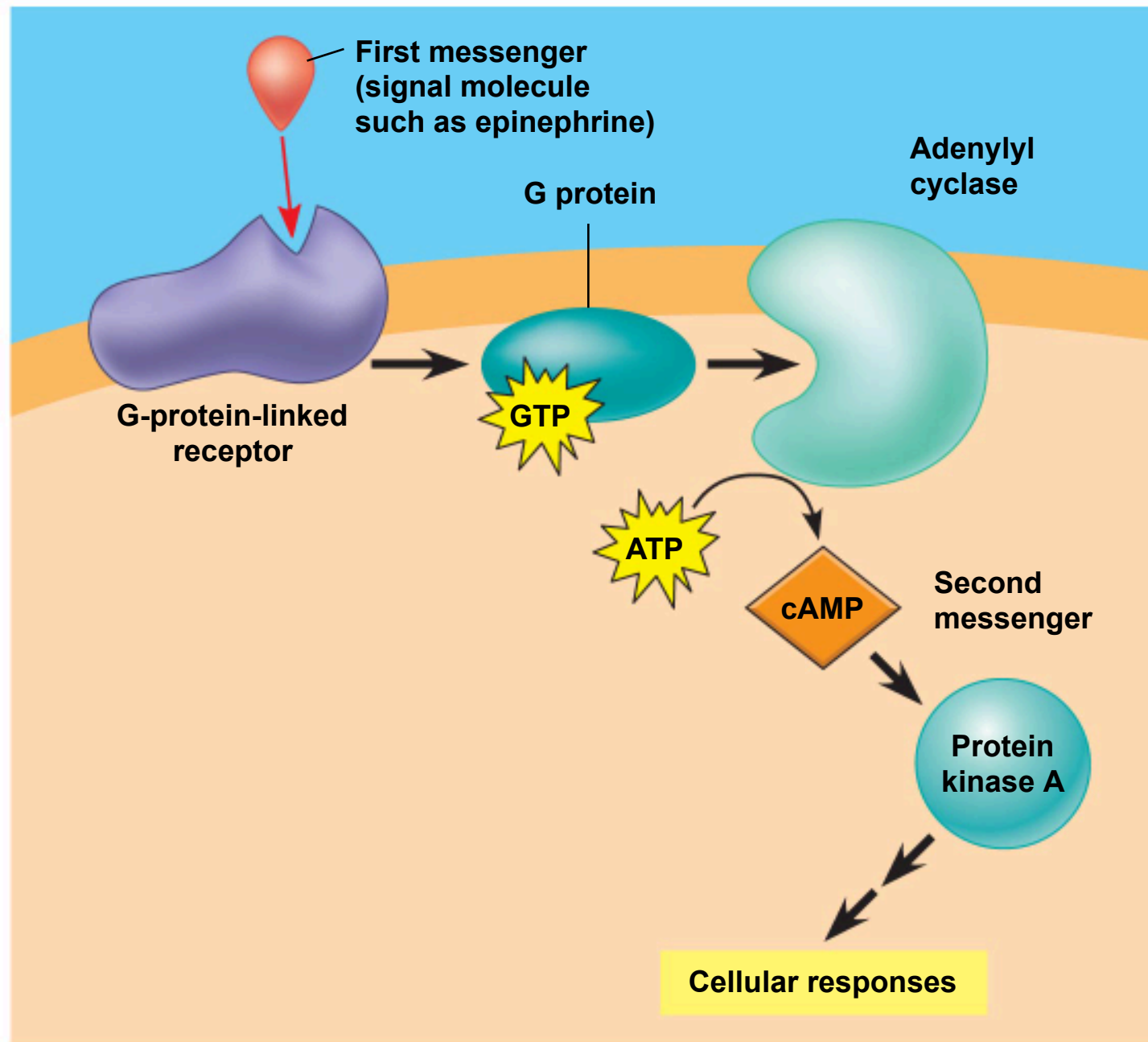
Small Molecules and Ions as Second Messengers

- Second messengers are small, nonprotein, water-soluble molecules or ions
- The extracellular signal molecule that binds to the membrane is a pathway's "first messenger"
- Second messengers can readily spread throughout cells by diffusion
- Second messengers participate in pathways initiated by G-protein-linked receptors and receptor tyrosine kinases
- Example: cAMP, Ca^{2+} , diacylglycerol

Cyclic AMP (cAMP)

- Cyclic AMP (cAMP) is one of the most widely used second messengers
- Adenylyl cyclase, an enzyme in the plasma membrane, converts ATP to cAMP in response to an extracellular signal

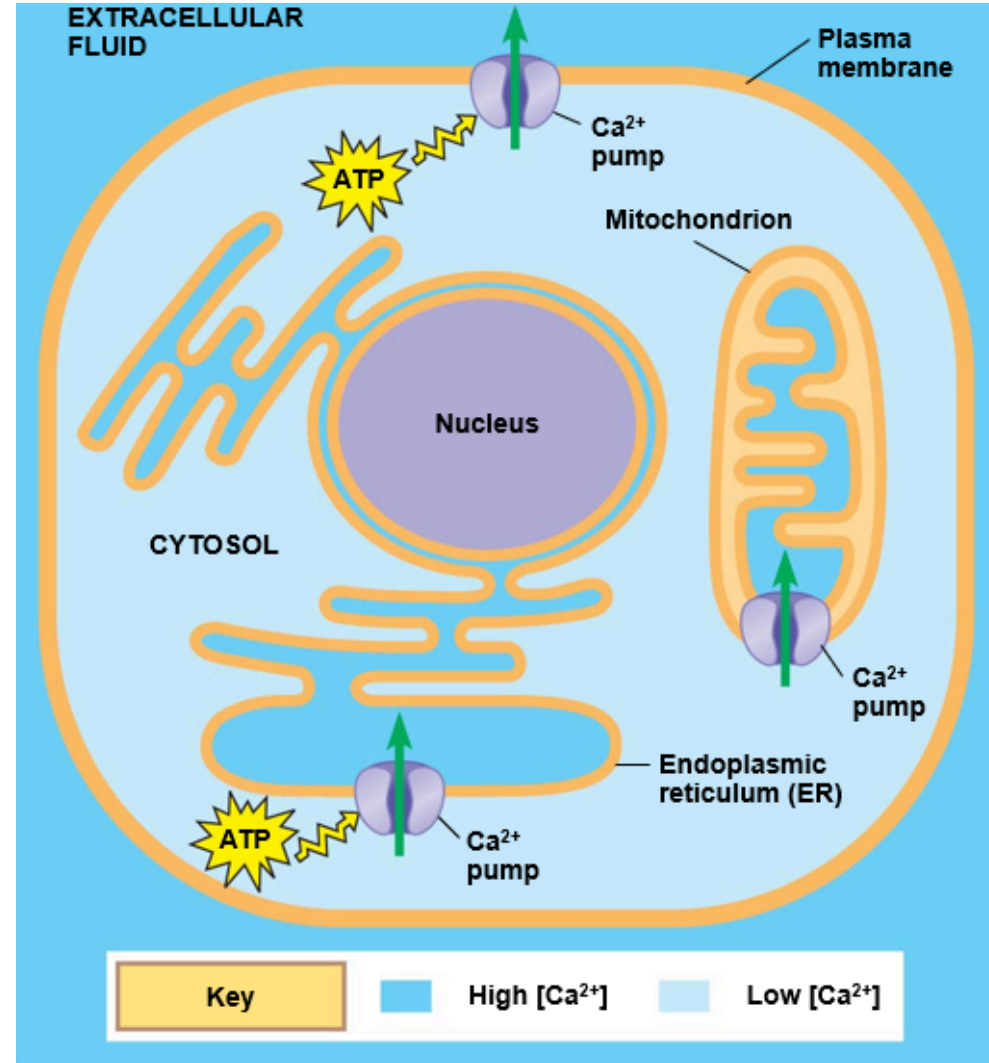




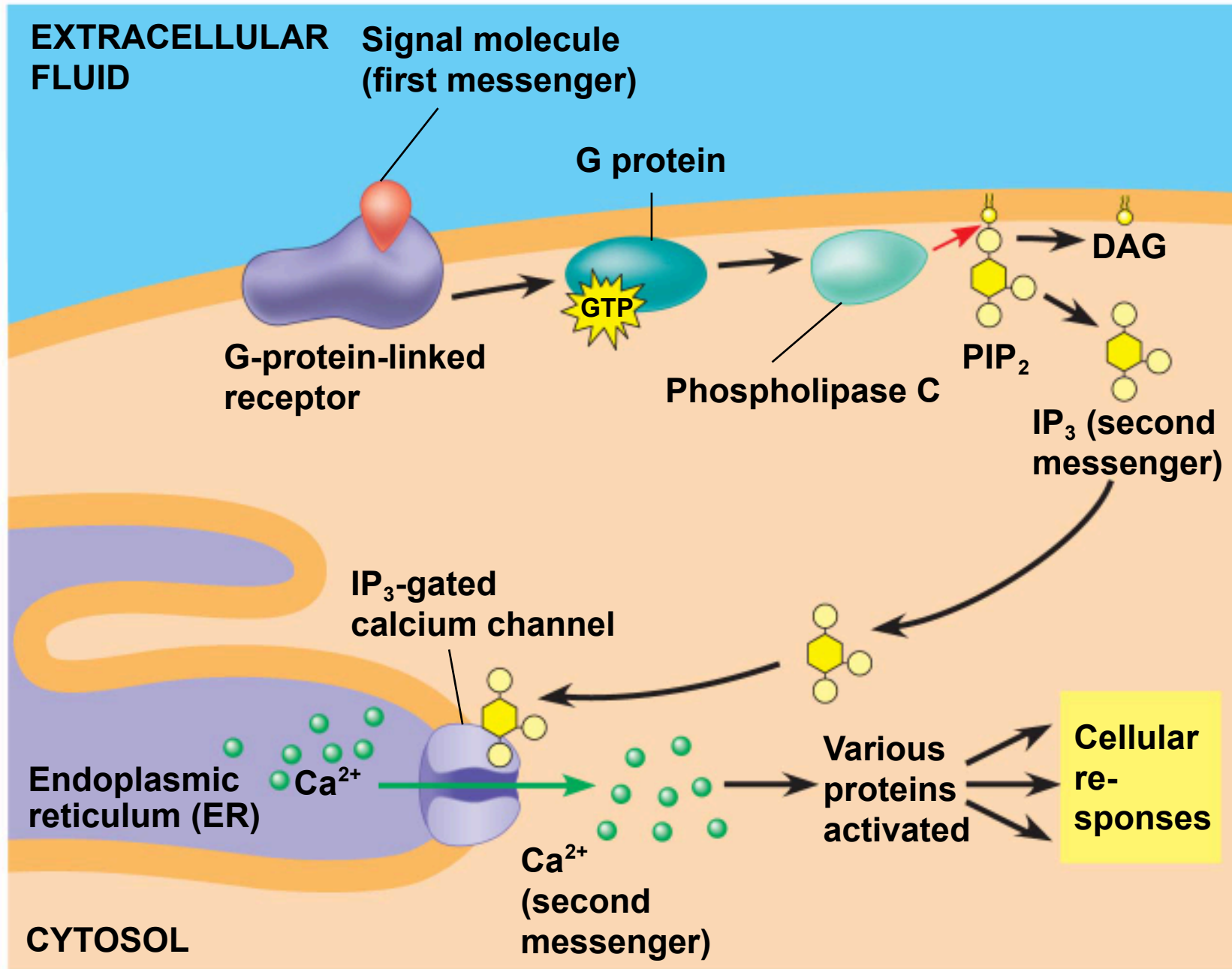
- Many signal molecules trigger the formation of cAMP
- Other components of cAMP pathways are G-protein, G-protein-linked receptor, and protein kinases
- cAMP usually activates protein kinase A, which phosphorylates various other proteins
- Further regulation of cell metabolism is G-protein system that inhibits adenylyl cyclase

Calcium ions and Inositol Triphosphate (IP_3)

- Calcium ions (Ca^{2+}) act as a second messenger in many pathways
- Calcium is an important second messenger because cells can regulate its concentration



- A signal relayed by a signal transduction pathway may trigger an increase in calcium in the cytosol
- Pathways leading to the release of calcium involve inositol triphosphate (IP_3) and diacylglycerol (DAG) as second messengers



CYTOPLASMIC AND NUCLEAR RESPONSE

- Cell signaling leads to the regulation of cytoplasmic activity or transcription
- Ultimately, a signal transduction pathway leads to regulation of one or more cellular activities
- The response may occur in the cytoplasm or in the nucleus
- Many signaling pathways regulate the synthesis of enzymes or other proteins, usually by turning genes on or off in the nucleus
- The final activated molecule in the signaling pathway may function as a transcription factor

Reception

Binding of epinephrine to G protein-coupled receptor (1 molecule)



Transduction

Inactive G protein

Active G protein (10^2 molecules)

Inactive adenylyl cyclase

Active adenylyl cyclase (10^2)

ATP

Cyclic AMP (10^4)

Inactive protein kinase A

Active protein kinase A (10^4)

Inactive phosphorylase kinase

Active phosphorylase kinase (10^5)

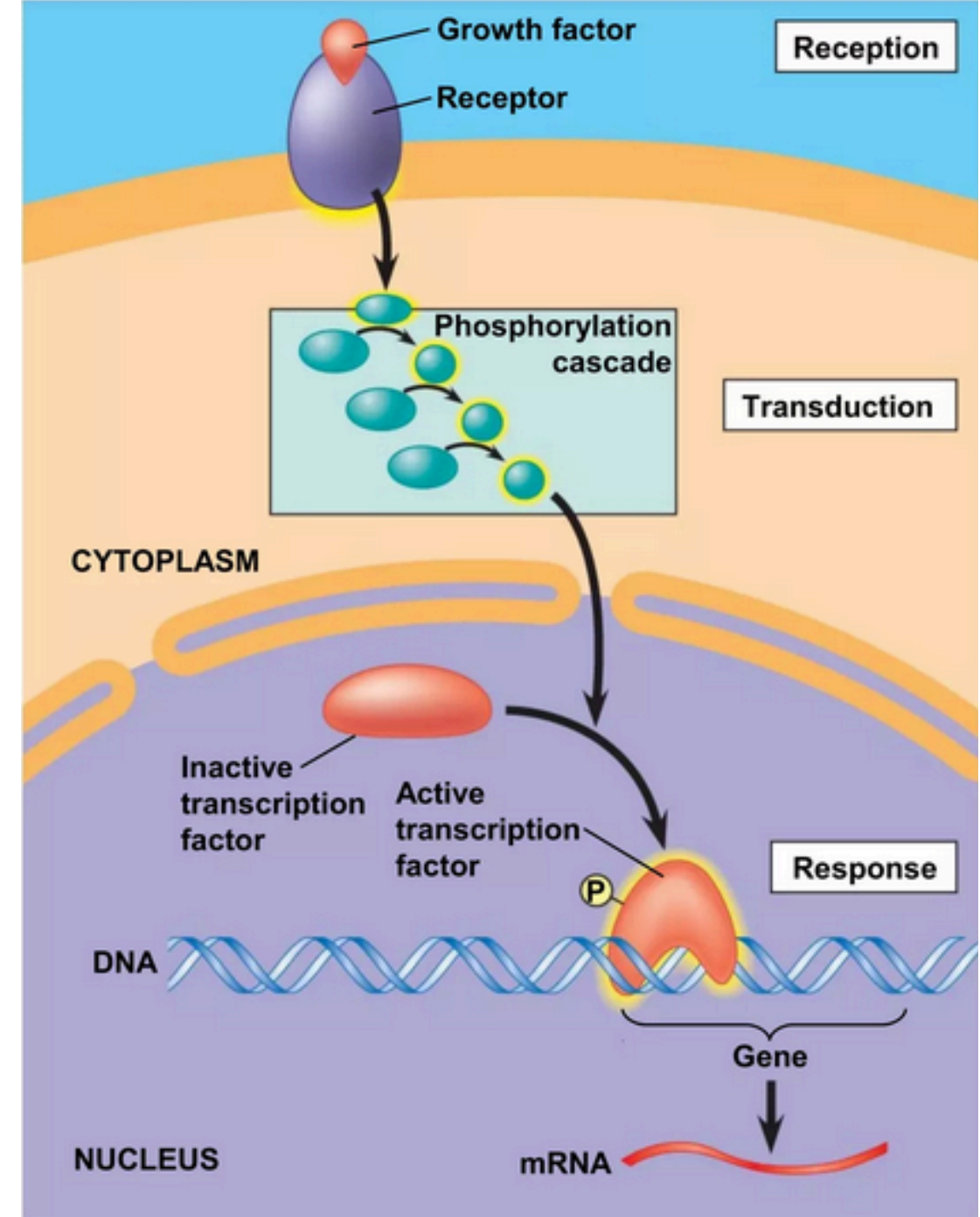
Inactive glycogen phosphorylase

Active glycogen phosphorylase (10^6)

Response

Glycogen
Glucose 1-phosphate
(10^8 molecules)

- Many other signaling pathways regulate the synthesis of enzyme or other protein, usually by turning genes on or off in the nucleus
- The final activated molecule may function as transcription factor



Fine-Tuning of the Response

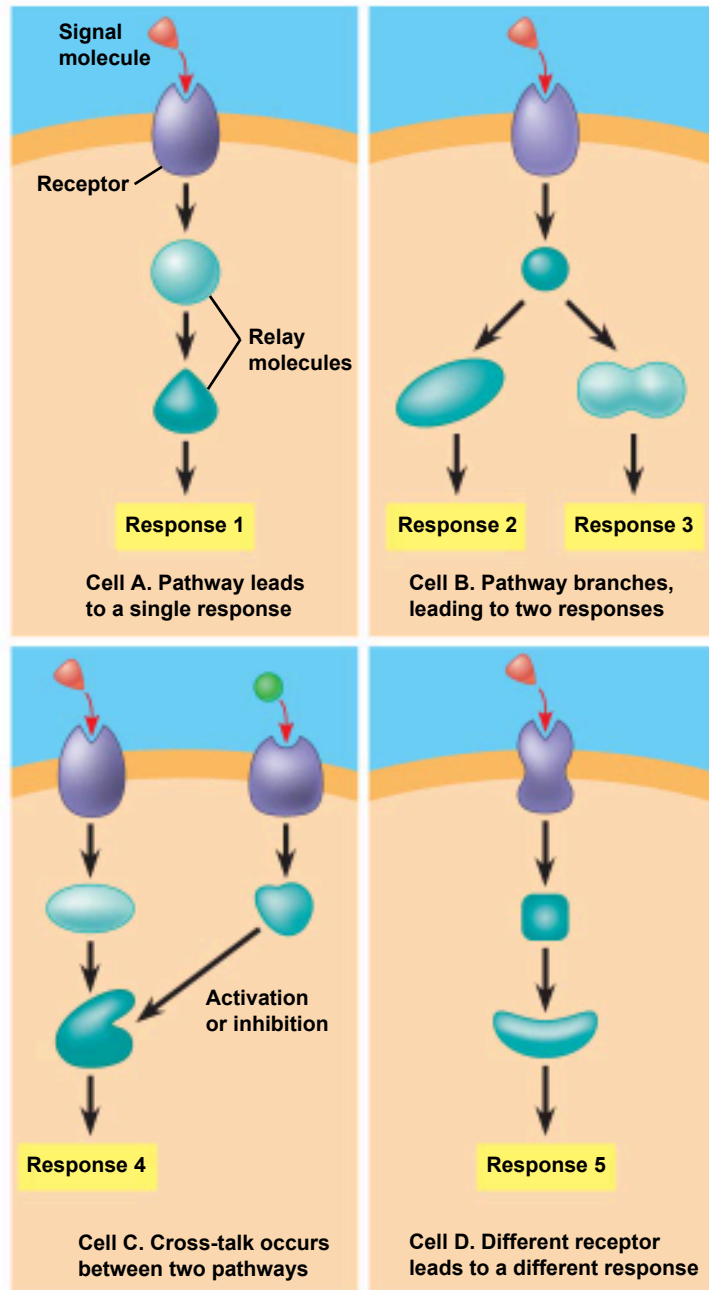
- There are four aspects of fine-tuning to consider
 - Amplifying the signal (and thus the response)
 - Specificity of the response
 - Overall efficiency of response, enhanced by scaffolding proteins
 - Termination of the signal

Signal Amplification

- Enzyme cascades amplify the cell's response
- At each step, the number of activated products is much greater than in the preceding step

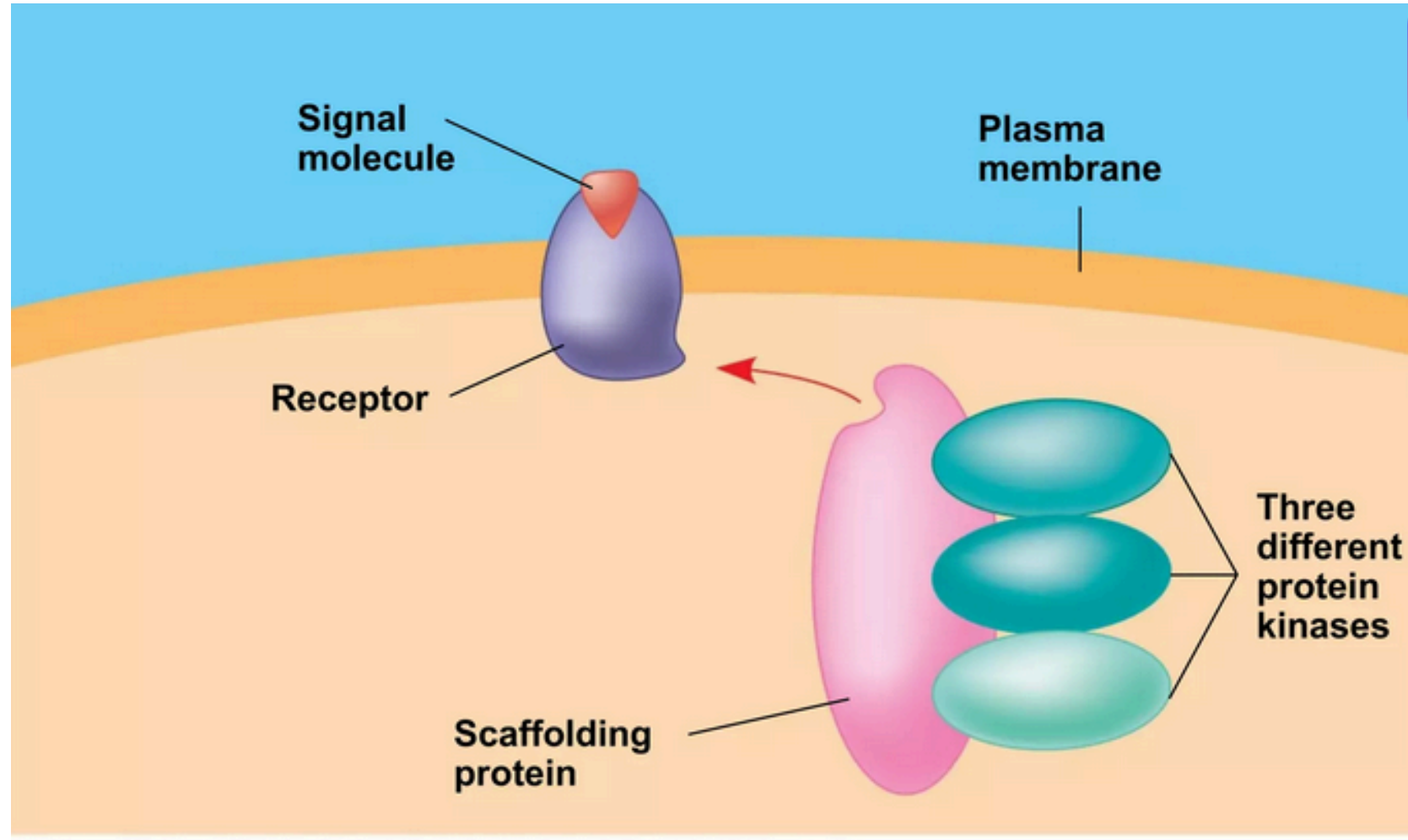
The Specificity of Cell Signaling

- Different kinds of cells have different collections of proteins
- These differences in proteins give each kind of cell specificity in detecting and responding to signals
- The response of a cell to a signal depends on the cell's particular collection of proteins
- Pathway branching and “cross-talk” further help the cell coordinate incoming signals



Signaling efficiency: Scaffolding protein

- Scaffolding proteins are large relay protein to which other relay proteins attached
- Scaffolding protein can increase the signal transduction efficiency



Termination of the Signal

- Inactivation mechanisms are an essential aspect of cell signaling
- When signal molecules leave the receptor, the receptor reverts to its inactive state

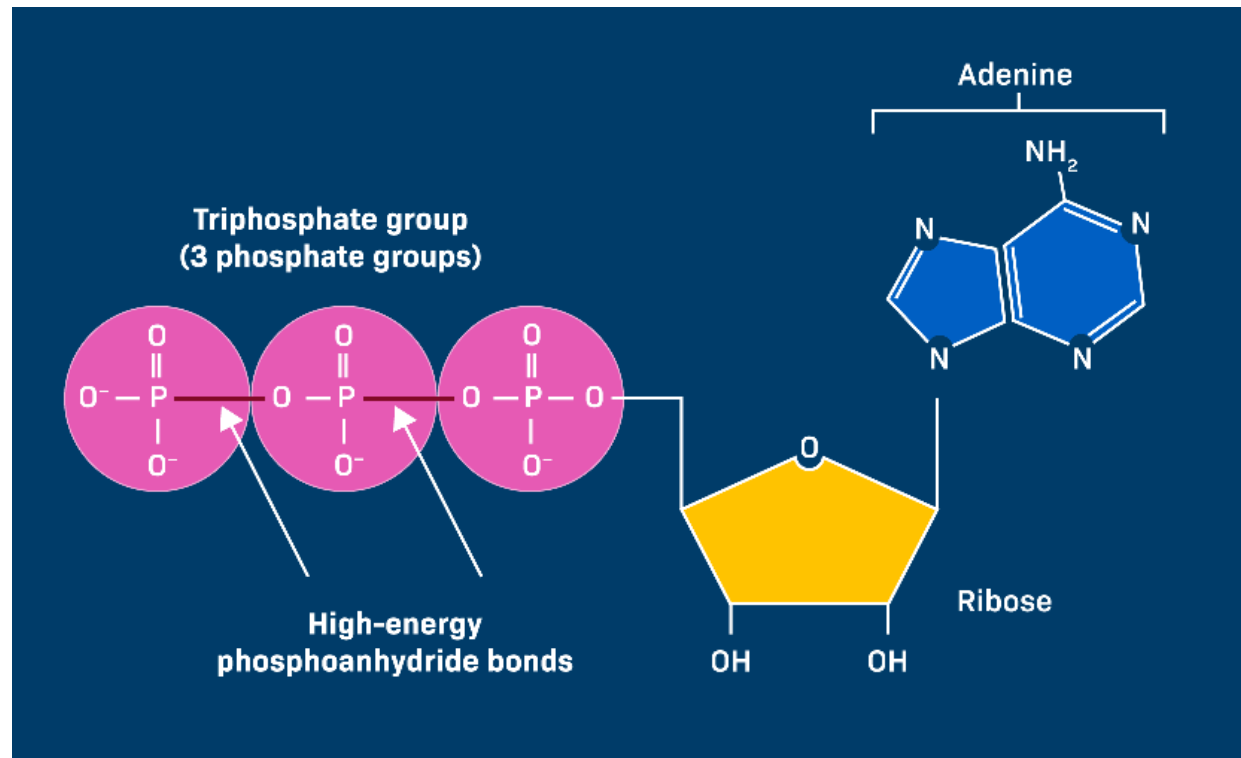
BIOENERGETICS

Bioenergetics Introduction

- Study of energy flow and transformation in living organisms
- Energy is essential for life processes
- Involves chemical reactions and energy transfer

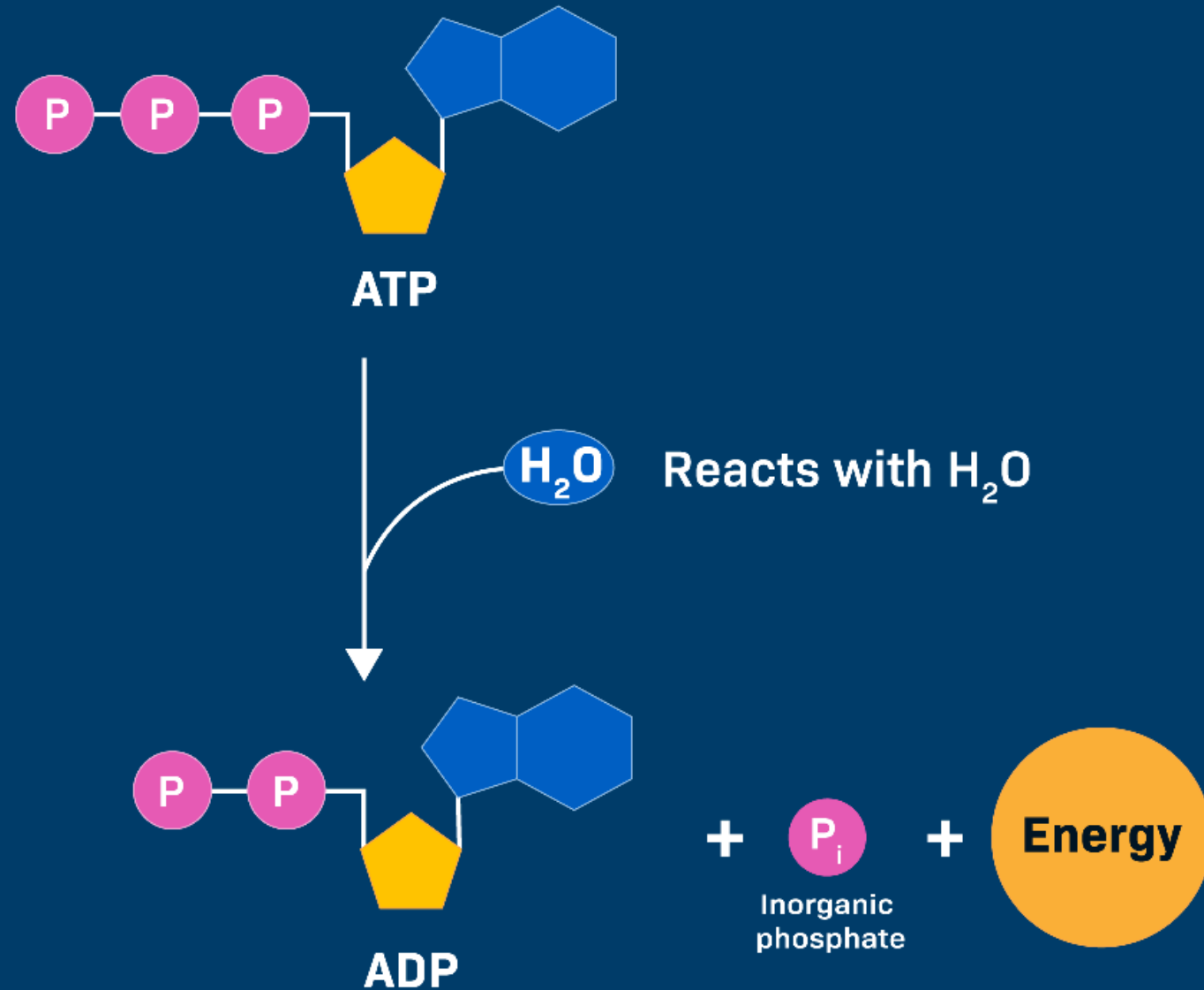
Adenosine Triphosphate (ATP): The Energy Currency

- Adenosine Triphosphate (ATP) powers cellular processes, driving processes like muscle contraction, active transport, and biosynthesis
 - Chemical energy is stored between its phosphate groups.



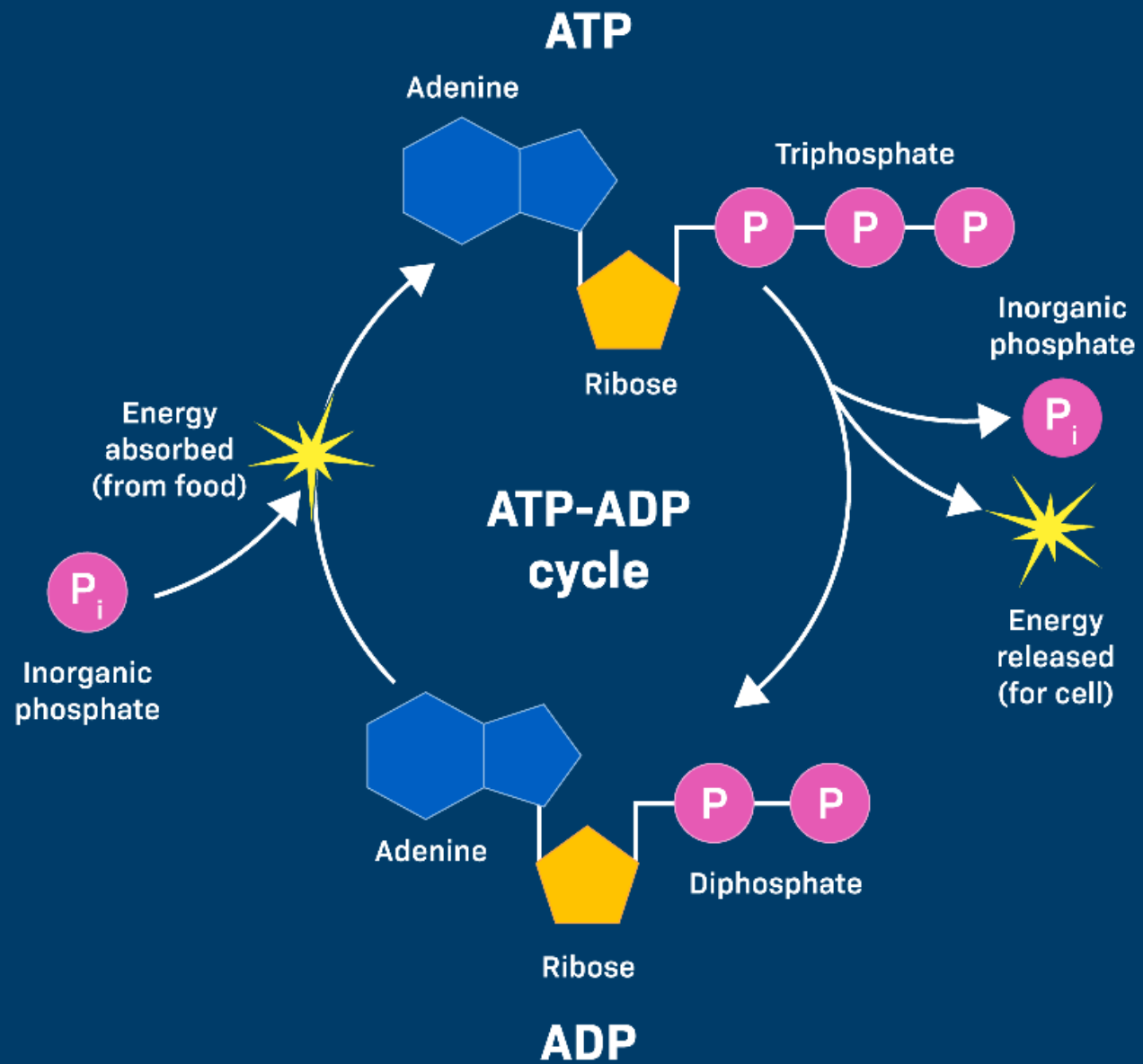
ATP – Adenosine Diphosphate (ADP) cycle

- ATP can be hydrolyzed by the addition of water, which breaks the bonds between the phosphate groups → exergonic reaction



ATP Regeneration:

- $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$, requiring energy input (endergonic reactions)
- Occurs via:
 - Oxidative Phosphorylation: In mitochondria, the electron transport chain (ETC) and ATP synthase use energy from glucose oxidation to produce ATP (~30-32 ATP per glucose).
 - Substrate-Level Phosphorylation: Direct transfer of phosphate to ADP in glycolysis and Krebs cycle (e.g., 2 ATP per glucose in glycolysis).



Adenosine Monophosphate (AMP)

- ADP can be further hydrolyzed: $\text{ADP} + \text{H}_2\text{O} \rightarrow \text{AMP} + \text{P}_i$, releasing additional energy.
- Alternatively, two ADP molecules combine: $2 \text{ADP} \rightarrow \text{ATP} + \text{AMP}$ (catalyzed by adenylate kinase), balancing energy levels in low-ATP conditions.
- Signaling: AMP levels rise when energy is low, activating pathways like AMP-activated protein kinase (AMPK), which stimulates energy-producing processes (e.g., glucose uptake, fatty acid oxidation).
- Reconversion: AMP can be rephosphorylated to ADP via adenylate kinase or other pathways, then to ATP.

Regulation

- Feedback Mechanisms: High ATP levels inhibit energy-producing pathways (e.g., glycolysis), while high AMP/ADP signals low energy, stimulating ATP production.
- Cellular Context: ATP/ADP/AMP ratios vary by cell type (e.g., high ATP in resting cells, high AMP in stressed cells)

Further reading

- Alberts B, Johnson A, Lewis J, Raff M, Roberts, Walker P. 2022. *The Molecular Biology of The Cell 7th ed*. Garland Science. UK

..... *Thank you*